Homework 3

Statistical Pattern Recognition

ℹ Instructions

Modify this quarto notebook, convert it to PDF using typst or to HTML and submit. Total is 70 points

{{< downloadthis week-03.qmd dname="homework3.qmd" label="Download Quarto file" >}}

Rubric:

1. Textual answers will be graded holistically. 5 points for an accurate and thoughtful answer, 4 points for minor errors, 3 points for a reasonable attempt with minor conceptual issues, 2 or less otherwise
2. Coding answers: 5 points for correct results, 4 points for correct approach, wrong answer, 3 points for an attempt which is mostly complete but got stuck in the end, 2 or less otherwise

Question 1: Conceptual – Supervised vs Unsupervised Learning (10 points)

(a)Explain the difference between supervised and unsupervised learning in the context of statistical pattern recognition.

(b) Provide one real-world data example from environmental or political science where supervised methods would be more appropriate, and one where unsupervised methods would be more useful. Justify your choices. Give examples not given in class 😃.

Question 2: Conceptual – Bias–Variance Tradeoff (15 points)

Consider the bias–variance tradeoff in the performance of pattern recognition models.

(a) Define bias and variance in this context. (5 points)

(b) Explain how increasing the complexity of a model (e.g., adding more predictors or using a flexible method like random forests) may affect bias and variance. (5 points)

(c) Simple models tend to be high-bias models, but have the advantage of not overfitting the training data. Give one scenario where a high-bias model might actually be preferable. (5 points)

Question 3: Data-Driven (Supervised Classification) (20 points)

You are given a dataset of air quality measurements (`airquality` in R). Suppose you want to build a supervised model to predict whether ozone concentration is above or below the median.

- Split the data into training (70%) and test (30%) sets. (2 points)
- Create a new binary variable `HighOzone` = 1 if `Ozone > median(Ozone, na.rm=TRUE)`, else 0. (3 points)
- Fit a logistic regression using `Solar.R`, `Wind`, and `Temp` as predictors. (5 points)
- Report the accuracy on the test set. (5 points)
- Explain why creating a dichotomous outcome variable with the median might not be desirable. What might be a possble method that might be preferred? (Extra credit, 5 points)

Some sample code to get you started:

```
# Install and load necessary libraries. Make sure you install the packages
separately from this document,
# so that you can run this
library(tidyverse)
library(tidymodels)
library(janitor)
library(datasets)

# Load data and remove missing values
data("airquality")
aq <- drop_na(airquality) # complete cases only

# Create binary outcome variable
median_ozone <- median(aq$Ozone)
aq$HighOzone <- factor(ifelse(________ , 1, 0))

# Split into training and test sets
set.seed(123)
splits<- initial_split(aq, prop = ____)
train <- analysis(splits) # the training data
test <- assessment(splits) # the test/evaluation data

# Fit logistic regression model
model <- glm(______ ~ ____________________,  data = train, family =
"binomial")
```

```r
# Predict and compute accuracy
test_performance <- augment(model, newdata = test, type.predict = "response")

## Create a new variable for predicted class based on a threshold of 0.5
# Look at names(test_performance) before to find the right variable name
test_performance <- mutate(test_performance,
                           pred_class = factor(ifelse(__________, 1, 0)),
                           HighOzone = as.factor(HighOzone))

# Look at the documentation for accuracy (?yardstick::accuracy) to figure out
how to calculate accuracy
acc <- accuracy(______, truth = ________, estimate = ________)

# Print accuracy
print(accuracy)
```

For binary classification using logistic regression, you actually get a probability score (between 0 and 1) for each observation. We will often try to figure out the best threshold for converting these probabilities into class labels (0 or 1). A common choice is 0.5, but this may not always be optimal depending on the context and costs of misclassification. We will often use ROC curves and AUC to evaluate classification models.

Implementing ROC curves (5 points)

You can read more about ROC curves and AUC here: https://www.statology.org/interpret-roc-curve

Now, reproduce this graph using ggplot2

Question 4: Unsupervised Clustering (25 points)

Using the built-in `iris` dataset:

- Perform a k-means clustering with `k = 3` using the numeric features (`Sepal.Length`, `Sepal.Width`, `Petal.Length`, `Petal.Width`). (5 points)
- Compare the resulting cluster assignments with the actual species labels. (5 points)
- Assign each cluster assignment to the species for which that cluster assignment is most common (5 points)
- Calculate and report the misclassification rate, i.e. the proportion of observations where the true species and the cluster assignment don't match (5 points)
- Briefly interpret whether k-means was effective in uncovering the underlying patterns without supervision. (5 points)